

GRIMSEL-HOSPIZ, Switzerland

WMO: 067440

Lat: 46.5717N

Lon: 8.3333E

Elev: 1965

StdP: 79.84

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.9	-13.8	-26.1	0.4	-9.1	-23.6	0.6	-8.6	18.8	-3.4	16.8	-1.8	3.8	350	0.847

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	19.1	11.9	17.7	11.3	16.3	10.7	12.9	17.1	12.1	16.3	11.4	15.3	4.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	10.7	14.0	10.4	10.0	13.3	9.5	9.4	12.8	42.7	17.2	40.4	16.4	38.3	15.5	16.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.5	10.2	DB	-18.4	21.8	3.3	1.2	-20.8	22.6	-22.7	23.3	-24.6	24.0	-26.9	24.9	(4)
(5)			WB	-19.1	14.3	3.1	1.0	-21.4	15.0	-23.2	15.6	-24.9	16.2	-27.2	16.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-4.9	-5.3	-2.6	0.9	4.3	9.1	10.7	10.5	7.2	4.8	-0.3	-3.5		
	DBStd	7.03	4.08	4.84	4.23	3.84	3.95	4.12	3.58	3.72	3.81	4.29	4.45	4.29		
	HDD10.0	2861	461	428	390	273	180	65	37	40	96	164	309	418		
	HDD18.3	5731	719	662	648	523	434	278	238	242	333	419	559	676		
	CDD10.0	172	0	0	0	0	4	38	57	56	13	4	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	1	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	5.3	5.9	5.8	5.6	5.3	4.9	4.6	4.6	4.8	4.6	5.5	5.8	5.9		
(15) Precipitation	PrecAvg	1579	98	104	103	125	161	154	140	154	138	127	160	115		
	PrecMax	2042	241	294	204	232	358	259	255	274	338	302	494	291		
	PrecMin	1106	24	20	19	20	45	60	75	48	41	15	31	12		
	PrecStd	208	58	63	53	57	80	49	46	48	70	68	101	61		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	6.6	7.5	11.5	16.4	21.3	20.7	21.0	17.1	15.0	9.9	5.8		
		MCWB	0.1	0.2	1.3	5.1	9.2	11.9	12.8	11.9	10.3	6.5	3.4	0.2		
	2%	DB	3.2	4.3	5.6	9.1	13.2	18.8	18.7	18.9	15.2	12.8	8.1	4.2		
		MCWB	-0.9	-1.3	0.3	3.9	7.4	11.5	12.1	11.9	9.4	6.1	2.6	-1.0		
	5%	DB	1.9	2.2	4.2	7.5	11.4	17.0	17.4	17.2	13.5	11.1	6.5	3.0		
		MCWB	-1.8	-2.0	-0.2	2.9	6.6	10.9	11.4	11.1	8.6	5.5	1.5	-1.3		
	10%	DB	0.5	0.5	2.9	6.0	9.7	15.1	16.0	15.8	12.2	9.8	5.1	1.9		
		MCWB	-2.2	-2.6	-0.8	2.2	5.6	10.1	10.9	10.6	7.9	4.9	1.2	-1.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	1.4	2.6	6.0	10.1	13.7	13.9	13.9	11.6	8.4	5.1	1.6		
		MCDB	3.4	4.2	5.1	9.7	14.5	17.1	18.2	17.7	14.6	12.3	8.5	3.6		
	2%	WB	0.4	0.0	1.5	4.6	8.3	12.5	12.9	12.7	10.4	7.4	3.6	0.8		
		MCDB	1.8	2.0	4.0	8.2	12.2	16.9	17.1	16.9	13.8	10.6	6.2	2.9		
	5%	WB	-0.8	-1.1	0.6	3.6	7.1	11.5	12.1	11.9	9.4	6.7	2.6	-0.1		
		MCDB	1.0	1.5	3.1	6.7	10.6	16.1	16.2	16.0	12.5	9.7	5.3	1.8		
	10%	WB	-1.8	-2.2	-0.2	2.6	6.1	10.5	11.3	11.1	8.5	5.9	1.8	-1.1		
		MCDB	0.1	0.4	2.0	5.4	9.2	14.5	15.1	14.8	11.3	9.0	4.5	1.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.1	5.4	5.3	5.6	6.3	6.0	5.8	5.3	5.2	4.4	4.7		
		MCDBR	5.5	6.2	6.4	6.6	7.6	8.7	7.7	7.4	6.8	6.9	5.6	5.1		
	5% WB	MCWBR	3.7	3.8	4.2	4.5	4.9	5.7	5.2	4.9	4.5	3.9	3.2	3.3		
		MCWBR	5.0	5.8	5.5	6.2	7.1	8.0	7.2	6.8	6.1	5.8	4.8	4.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.229	0.253	0.294	0.337	0.354	0.361	0.359	0.349	0.325	0.299	0.260	0.227		
		taud	2.339	2.283	2.233	2.227	2.286	2.313	2.311	2.356	2.417	2.470	2.454	2.370		
	Ebn at Noon	Ebn at Noon	894	931	931	917	910	902	898	895	890	868	848	856		
		Edh at Noon	71	94	117	130	128	125	124	113	97	79	64	61		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.54	2.50	3.69	4.44	4.82	5.67	5.64	4.90	3.82	2.52	1.53	1.28		
		RadStd	0.19	0.30	0.53	0.70	0.47	0.51	0.49	0.44	0.37	0.29	0.22	0.17		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (118 neighbors)	N/A	N/A	N/A	+0.73	+0.42	+0.32	N/A	N/A	+58	+16	(47)

Nomenclature: See separate page